

Core Pure Mathematics 1

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Chapter 7: Linear Transformations

Why this matters

This chapter is where matrices from Chapter 6 become geometrically meaningful. Every 2×2 (and 3×3) matrix can be interpreted as a transformation of the plane (or space): a rotation, reflection, enlargement, or stretch. This is a highly visual, intuitive topic once the link between algebra and geometry clicks — and it is a favourite source of exam questions combining matrix mechanics with geometric reasoning.

Concept: matrices as transformations

A linear transformation maps every point (x, y) to a new point according to $(x'; y') = M(x; y)$ for some matrix M . Because matrix multiplication is linear, straight lines map to straight lines and the origin always maps to itself. The columns of M tell you everything: the first column is where $(1, 0)$ maps to, and the second column is where $(0, 1)$ maps to.

The determinant of M has a direct geometric meaning too: $|\det(M)|$ is the factor by which M scales area (or volume, in 3D). This is why a singular matrix ($\det(M) = 0$, from Chapter 6) squashes the whole plane down to a line or a point — it scales every area down to zero. A negative determinant scales area by $|\det(M)|$ as usual, but also reverses orientation (flips the plane over, as a reflection does).

Key vocabulary

Term	Meaning
Linear transformation	A mapping of the plane (or space) representable by matrix multiplication
Image	The point or shape a transformation maps something to
Invariant point	A point that maps to itself under a transformation, i.e. $Mx = x$
Invariant line	A line that maps to itself as a whole (points on it may move along it)
Composite transformation	Applying one transformation after another; represented by multiplying matrices

Everything here extends naturally to 3×3 matrices acting on 3D space (rotations about an axis, reflections in a plane, and so on) — the same column-reading trick and the same rules for composing transformations still apply.

Concept: standard matrices — rotations, reflections, enlargements

Certain transformations have standard matrix forms that should be learned:

Rotation by angle θ anticlockwise about the origin: $(\cos\theta \ -\sin\theta; \sin\theta \ \cos\theta)$

Reflection in the x-axis: $(1 \ 0; 0 \ -1)$. Reflection in the y-axis: $(-1 \ 0; 0 \ 1)$. Reflection in the line $y = x$: $(0 \ 1; 1 \ 0)$

Enlargement, scale factor k , centre origin: $(k \ 0; 0 \ k)$

Stretch, scale factor k parallel to the x-axis: $(k \ 0; 0 \ 1)$. Stretch, scale factor k parallel to the y-axis: $(1 \ 0; 0 \ k)$

Worked example 1: finding the image of a shape under a transformation

EXAMPLE 1

Question. A triangle has vertices $A(1, 0)$, $B(2, 1)$, $C(0, 2)$. Find the image of the triangle under the transformation with matrix $M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and describe the transformation geometrically.

Step 1. Transform each vertex by multiplying M by its column vector. For A : $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, so A maps to $(0, 1)$. **(M1 A1)**

Step 2. For B : $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$, so B maps to $(-1, 2)$. **(A1)**

Step 3. For C : $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$, so C maps to $(-2, 0)$. **(A1)**

Step 4. Recognise M as the standard rotation matrix with $\cos\theta = 0$, $\sin\theta = 1$, i.e. $\theta = 90^\circ$. **(M1 A1)**

Answer: The image is the triangle with vertices $(0,1)$, $(-1,2)$, $(-2,0)$; M represents a rotation of 90° anticlockwise about the origin.

Exam tip

EXAM TIP

Identify a transformation matrix by asking "where does $(1,0)$ go, and where does $(0,1)$ go?" and comparing directly to the standard matrices, rather than trying to recall the standard matrices from memory alone. This column-by-column reading is a much more robust skill than memorisation.

Concept: combining transformations

If transformation P is applied first, followed by transformation Q , the combined (composite) transformation is represented by the matrix product QP — note the reversed order: the matrix for the transformation applied first goes on the right.

Worked example 2: composite transformation

EXAMPLE 2

Question. P is a reflection in the x-axis and Q is a rotation of 90° anticlockwise about the origin. Find the matrix representing " P followed by Q ", and use it to find the image of the point $(3, 2)$.

Step 1. Write down the matrices: $P = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. **(B1)**

Step 2. Since P is applied first, the combined matrix is QP (Q applied to the result of P): $QP = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. **(M1)**

Step 3. Multiply: top-left = $(0 \times 1) + (-1 \times 0) = 0$; top-right = $(0 \times 0) + (-1 \times -1) = 1$; bottom-left = $(1 \times 1) + (0 \times 0) = 1$; bottom-right = $(1 \times 0) + (0 \times -1) = 0$. So $QP = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. **(M1 A1)**

Step 4. Apply QP to $(3, 2)$: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. **(M1 A1)**

Answer: The combined matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (a reflection in the line $y = x$), and it maps $(3, 2)$ to $(2, 3)$.

Exam tip

EXAM TIP

The order-reversal rule ("P then Q" gives matrix QP , not PQ) trips up almost everyone at first. A reliable fix: test a single point through each transformation step by step **BEFORE** multiplying the matrices, then check that QP applied directly to the same point gives the same answer — this catches order mistakes immediately rather than after the exam.

Concept: invariant points and invariant lines

A point x is **invariant** if it maps to itself: $Mx = x$, i.e. $Mx - x = 0$, i.e. $(M - I)x = 0$. For a non-trivial (non-zero) invariant point to exist besides the origin, we need $\det(M - I) = 0$.

An **invariant line** is different — it's a line mapped onto itself as a whole, though individual points on it may slide along it. To find one: assume a general point on a line $y = mx + c$, transform it with M , and require the image to also satisfy the same line equation.

Worked example 3: finding an invariant line

EXAMPLE 3

Question. Find the invariant lines through the origin for the transformation with matrix $M = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$.

Step 1. Consider a line through the origin, $y = mx$. A general point on it is (t, mt) for some t . **(M1)**

Step 2. Transform this point: $M(t; mt) = (3t + 2mt, 2t + 3mt) = (t(3 + 2m), t(2 + 3m))$. **(M1 A1)**

Step 3. For the image to lie on the same line $y = mx$, we need: $t(2 + 3m) = m \times t(3 + 2m)$. **(M1)**

Step 4. Divide by t ($t \neq 0$) and simplify: $2 + 3m = 3m + 2m^2$, so $2 = 2m^2$, giving $m^2 = 1$, so $m = 1$ or $m = -1$. **(M1 A1)**

Answer: The invariant lines through the origin are $y = x$ and $y = -x$.

Exam tip

EXAM TIP

It's worth knowing that invariant lines through the origin correspond to eigenvectors of the matrix (a concept you may meet formally at university) — you don't need this language for the exam, but if you are heading toward further study, the connection adds valuable context and motivation.

Chapter summary

Idea	Key point
Reading a matrix	Column 1 = image of (1,0); column 2 = image of (0,1)
Standard matrices	Learn rotation, reflection and enlargement/stretch matrices
Composite transformations	"P then Q" is represented by QP (reversed order)
Invariant points	$Mx = x$, i.e. $(M - I)x = 0$; non-zero solutions need $\det(M - I) = 0$
Invariant lines	A line mapped onto itself; find by substituting $y = mx (+c)$ and matching coefficients

Common exam mistakes

- Multiplying transformation matrices in the wrong order for a composite transformation.
- Confusing an invariant point (maps to itself exactly) with an invariant line (maps onto itself, but points can move along it).
- Forgetting that a stretch/enlargement matrix must be diagonal, and mixing up which axis a stretch factor applies to.
- Sign errors in the rotation matrix, especially for clockwise rotations (angle should be taken as negative).
- Forgetting to divide by t (or otherwise handle it correctly) when deriving invariant line conditions, or losing solutions by dividing by an expression that could be zero.

You should now be able to

- Interpret a matrix as a geometric transformation and find the image of points or shapes under it.
- Recall and apply the standard matrices for rotations, reflections, enlargements and stretches.
- Find the matrix representing a composite of two or more transformations, in the correct order.
- Find invariant points and invariant lines for a given transformation.